DualNet-AL: A Bi-Network Meta-Learning Framework for Multi-Class Active Learning

Himansh Raj, Om Narayan Pandey, and Jaswinder Singh
School of Computer Science & Engineering, IILM University, Greater Noida, India
{himansh.raj.cs27, om.pandey.cs27, jaswinder.singh}@iilm.edu

***Abstract*—Pool-based active learning reduces annotation cost by letting a model choose which unlabelled images to label, but single-criterion strategies are brittle: uncertainty sampling picks redundant or outlying samples, a single model reinforces its own errors, and at small budgets the model's uncertainty is itself unreliable. We present DualNet-AL, which trains two architecturally different convolutional networks, scores every unlabelled sample with five complementary signals (uncertainty, inter-network disagreement, representativeness, diversity and class balance), fuses them with weights adapted online by a meta-learner, and, in its second version (v2), combines a budget-aware prior that favours typical samples while labels are scarce with cluster-based batch construction. Configuration was chosen on validation data only and evaluated on held-out seeds and the official test splits. Across Fashion-MNIST, SVHN and CIFAR-10 (100→900 labels), DualNet-AL v2 improves the area under the learning curve over random sampling by +2.41, +4.38 and +1.46 percentage points, winning on every one of the 3 held-out seeds of each dataset, is the best of nine methods on SVHN and CIFAR-10, and more than doubles the worst-class recall of random sampling on SVHN (40.5% vs 16.7%).**

***Index Terms*—active learning, deep learning, meta-learning, query by committee, label efficiency, image classification.**

I. Introduction

Supervised deep networks need many labelled examples, and labelling is the dominant cost in most applied projects. Active learning (AL) addresses this by training on a small labelled set and repeatedly asking an oracle to label the samples the current model considers most useful [1]. Classical acquisition functions rank samples by a single criterion—predictive entropy [2], the margin between the two most likely classes [3], disagreement within a committee [4], or mutual information under model uncertainty (BALD) [5], [6]. Diversity-based methods instead choose samples that cover the feature space, e.g. the core-set approach [7].

Each criterion fails in a predictable way. Uncertainty-driven batches are redundant and drawn towards outliers; a single model choosing its own training data reinforces its own mistakes; and, as recently shown, at low budgets uncertainty is a poor guide and typical samples should be preferred instead [10]. The best criterion therefore depends on the dataset and on how many labels have been collected.

DualNet-AL tackles these failure modes together. Our contributions are: (i) a bi-network learner whose two members differ in architecture and normalisation, so that their disagreement is an informative, committee-style signal; (ii) five rank-normalised acquisition signals fused by a meta-learner that re-weights them every round from measured validation gain and from label-revealed errors; (iii) a v2 selection rule that adds a budget-aware typicality prior and one-sample-per-cluster batches; and (iv) a leak-free evaluation of nine strategies under identical networks and compute, with all code and per-run logs released at https://github.com/iamthehimansh/DualNet-AL_Motion.

II. Related Work

Uncertainty and committees. Entropy [2] and margin [3] sampling are strong, cheap baselines; query by committee [4] measures disagreement among several models, and BALD [5], [6] approximates a Bayesian committee with Monte-Carlo dropout. Co-training [22] showed the value of two different views of the data, which motivates our use of two architecturally different networks.

Diversity and batch construction. Core-set selection [7] picks a k-center cover of the embedding space; BADGE [8] combines uncertainty and diversity in gradient space; pre-clustering [13] selects representatives of clusters. Learning-loss [9] trains an auxiliary loss predictor.

Learned and budget-aware strategies. Konyushkova et al. [11] learn an acquisition function from data and Hsu and Lin [12] select among strategies with a bandit. Hacohen et al. [10] show that typical samples help at low budgets while uncertain ones help at high budgets. DualNet-AL combines these ideas: a bandit-style meta-learner over signals rather than whole strategies, and an explicit schedule from typicality to informativeness.

III. Method

A. Problem setting

We start from a labelled set L of 100 random samples and an unlabelled pool U. In each of R=8 rounds a batch of B=100 samples is selected from U, labelled by the oracle and moved to L, after which the model is retrained. The goal is maximum test accuracy for every budget.

B. Bi-network learner

Net A is a 6-layer 3×3 CNN with BatchNorm and max-pooling (289k parameters for RGB input); Net B is a wide 5×5 CNN with GroupNorm, LeakyReLU, average pooling and an MLP head (169k parameters). Both are retrained from scratch every round for a fixed 300 AdamW steps, and predictions are the average of their softmax outputs. Their concatenated penultimate features form the embedding used by the representativeness, diversity and clustering steps.

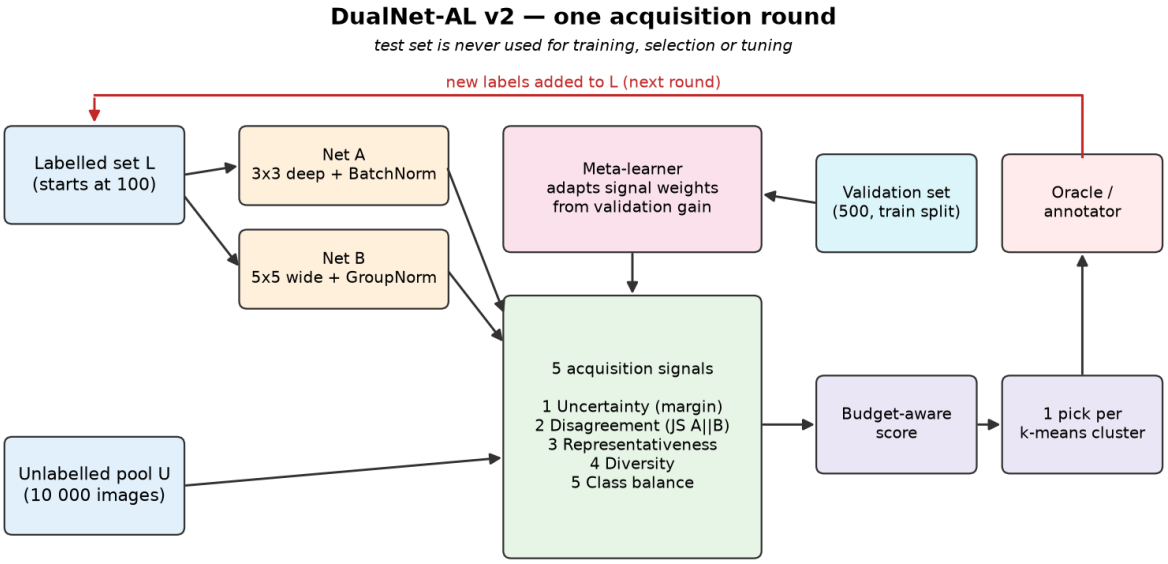


Fig. 1. DualNet-AL v2: one acquisition round. The test set is never used for training, selection or tuning.

C. Acquisition signals

For every pool sample we compute five scores, each mapped to [0,1] by rank-percentile normalisation so that their very different raw scales become comparable: **uncertainty** $s_1=1-(\bar{p}_{(1)}-\bar{p}_{(2)})$, the margin of the averaged prediction $\bar{p}=(p_A+p_B)/2$ [3]; **disagreement** $s_2=0.7\cdot \mathrm{JS}(p_A\|p_B)/\log 2+0.3\cdot[\text{argmax } p_A \neq \text{argmax } p_B]$ using the Jensen–Shannon divergence [14]; **representativeness** s_3, the mean cosine similarity to the 10 nearest pool neighbours; **diversity** s_4, the distance to the closest labelled sample [7]; and **class balance** s_5, the rarity of the predicted class in L.

D. Meta-learned weighting

Signals are fused as $S(x)=\Sigma_k\ w_k s_k(x)$ with $w=w_{\min}+(1-5w_{\min})\cdot\mathrm{softmax}(\theta)$, $w_{\min}=0.05$, so no signal is ever switched off. After the batch chosen in round t has been labelled and the networks retrained, θ is updated by

$$\theta_k \leftarrow \theta_k + \eta\cdot a\cdot c_k + \eta_c\cdot\rho_k,$$

where a is the validation-accuracy gain normalised against a moving baseline (a REINFORCE-style advantage [15]), c_k is how much the batch relied on signal k (its batch mean minus its pool mean) and ρ_k is the correlation between signal k and the ensemble's errors on the newly labelled batch; $\eta=1.5$, $\eta_c=0.8$.

E. Budget-aware clustered selection (v2)

Version 1 ranked samples by S and removed near-duplicates. In v2, the score becomes

$$score(x) = (1-\lambda_t)\cdot s_3(x) + \lambda_t\cdot S(x),\quad \lambda_t = \lambda_0 + (1-\lambda_0)\cdot\min(1, t/r),$$

so early rounds favour typical samples [10] and the meta-learned score takes over after r rounds. The pool embedding is then partitioned into B clusters with cosine k-means (k-means++ seeding [16]) on the GPU, and the best-scoring sample of each cluster is selected, which guarantees coverage and removes redundancy [13]. The chosen setting is $\lambda_0=0.5$, $r=4$, margin uncertainty (Section IV-C).

Algorithm 1 DualNet-AL v2 (one round t)

1: train Net A, Net B on L; predict p_A, p_B and embeddings on U
2: compute and rank-normalise $s_{1..5}$; fuse S with meta weights w
3: score $\leftarrow (1-\lambda_t)s_3 + \lambda_t S$; k-means($U$, B)
4: batch $\leftarrow$ argmax score in each cluster; oracle labels batch; $L \leftarrow L \cup$ batch
5: after retraining: update θ from validation gain and batch errors

IV. Experimental Setup

A. Datasets and protocol

We use Fashion-MNIST [17], SVHN [19] and CIFAR-10 [18]. For each dataset and seed, a 10 000-image pool and a 500-image validation set are drawn from the official training split, and 2 000 test images from the official test split; inputs are standardised with pool statistics only. Every strategy starts from the same 100 random labels for a given seed, adds 100 labels per round for 8 rounds and uses the same two networks and the same training budget; only the selection rule differs. Accuracy is that of the A+B ensemble.

B. Baselines and metrics

We compare Random, Entropy [2], Margin [3], BALD with 10 MC-dropout passes [6], Core-Set k-center greedy [7], QBC over our two networks [4], and DualNet-AL v1 with fixed and with meta-learned weights. The main metric is the area under the accuracy-vs-labels curve (AULC), normalised by the label range; we also report final accuracy, macro-F1, worst-class recall, the number of labels needed to reach Random's final accuracy, selection time and expected calibration error (ECE) [20].

C. Tuning without test-set leakage

Six v2 configurations and two references were run on tuning seeds 0–2 with the test set disabled, scoring validation AULC only (Table III). A single configuration—the one with the largest mean gain over Random across the three datasets—was frozen and then evaluated, together with all baselines, on new seeds 3, 4, 5 on the test splits. Experiments used PyTorch [21]

2.11.0 on an NVIDIA GeForce RTX 5060 Ti; each run of 9 retraining rounds took under a minute.

V. Results

Table I reports held-out AULC. DualNet-AL v2 is the best of nine methods on SVHN and CIFAR-10 and ranks 2nd on Fashion-MNIST, where Margin is 0.17 pp higher. Table II gives the paired comparison with Random on identical seeds and starting labels: v2 is better on every seed of every dataset, by +2.41, +4.38 and +1.46 pp AULC on Fashion-MNIST, SVHN and CIFAR-10. It reaches Random's final (900-label) accuracy after about 346, 715 and 684 labels respectively.

TABLE I. Held-out AULC (%), mean ± std over seeds 3, 4, 5; best in bold.

Method	Fashion-MNIST	SVHN	CIFAR-10
Random	78.92 ± 1.18	53.17 ± 1.49	47.20 ± 0.30
Entropy [2]	77.86 ± 0.88	38.71 ± 1.57	45.88 ± 1.01
Margin [3]	**81.49 ± 0.38**	49.34 ± 2.70	47.95 ± 0.21
BALD [5,6]	79.94 ± 0.48	55.89 ± 1.25	47.05 ± 1.19
Core-Set [7]	78.42 ± 0.66	40.18 ± 1.65	46.81 ± 1.17
QBC [4]	79.63 ± 0.81	47.58 ± 1.77	46.30 ± 0.64
DualNet-AL v1 (fixed)	80.72 ± 0.82	51.08 ± 1.56	48.46 ± 0.34
DualNet-AL v1 (meta)	80.04 ± 0.67	49.30 ± 4.34	48.26 ± 0.68
DualNet-AL v2 (ours)	81.33 ± 0.34	**57.56 ± 1.12**	**48.66 ± 0.52**

TABLE II. DualNet-AL v2 vs Random on the same held-out seeds (v2 / Random).

Dataset	ΔAULC (pp)	Seeds better	Final acc.	Macro-F1	Worst-class rec.	Labels to Random@900
Fashion-MNIST	+2.41	3/3	83.65 / 81.02	83.51 / 80.72	46.8 / 38.0	346
SVHN	+4.38	3/3	69.70 / 70.20	65.77 / 63.91	40.5 / 16.7	715
CIFAR-10	+1.46	3/3	53.98 / 52.93	53.07 / 51.70	27.9 / 21.2	684

The uncertainty-only methods collapse on SVHN (Entropy 38.71%, Core-Set 40.18% vs Random 53.17%): with a few hundred labels their batches are dominated by confusing, redundant digits. DualNet-AL v1 also lost to Random there (49.30%), which motivated v2. The fairness benefit is large: worst-class recall rises from 16.7% to 40.5% on SVHN, from 38.0% to 46.8% on Fashion-MNIST and from 21.2% to 27.9% on CIFAR-10.

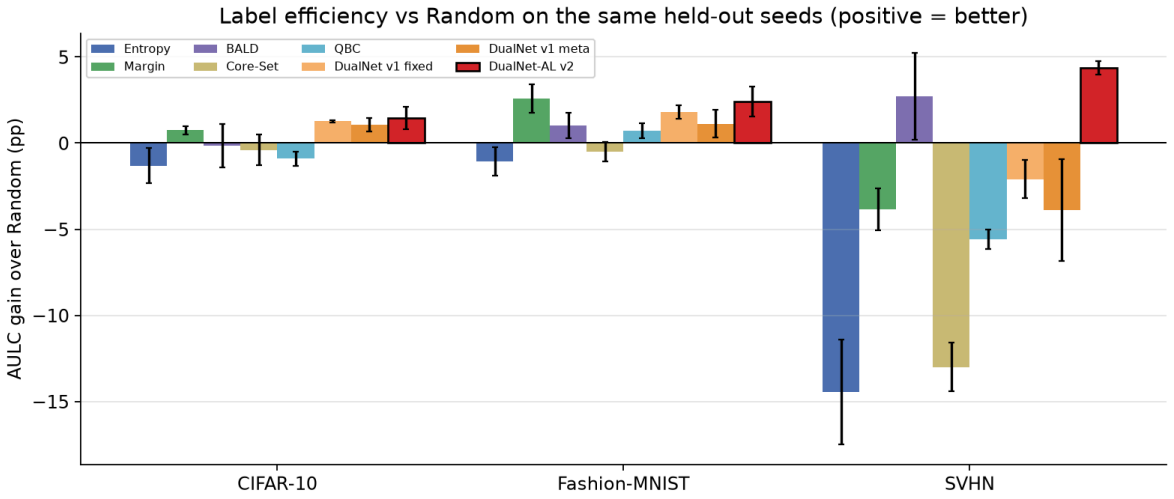


Fig. 2. AULC gain over Random on the same held-out seeds (error bars: std over seeds). Positive is better.

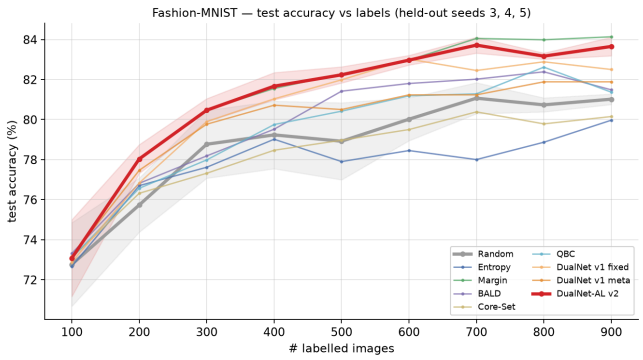


Fig. 3. Fashion-MNIST: test accuracy vs labels (mean ± std).

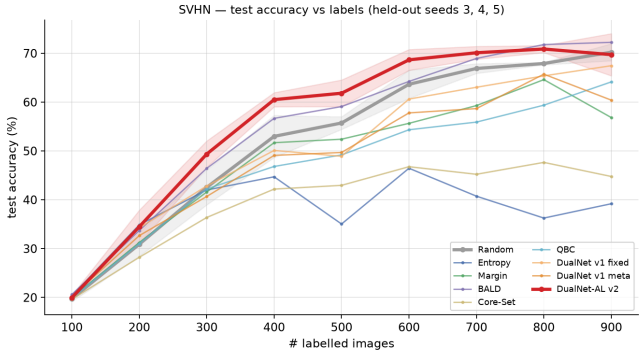


Fig. 4. SVHN: test accuracy vs labels (mean ± std).

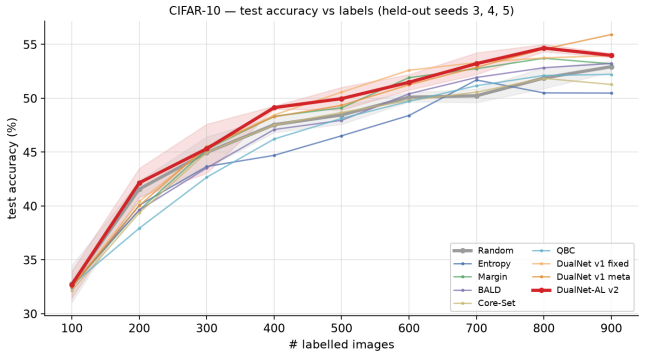


Fig. 5. CIFAR-10: test accuracy vs labels (mean ± std).

Ablation and tuning

Table III shows the validation results used for model selection. Removing the budget-aware prior (λ=1 throughout) costs 4.93 pp on SVHN and 1.90 pp on CIFAR-10 relative to the chosen setting (-0.08 pp on Fashion-MNIST), confirming that typical samples matter most while labels are scarce. With the same schedule, entropy instead of margin uncertainty changes validation AULC by -0.31, -0.26 and -0.65 pp on Fashion-MNIST, SVHN and CIFAR-10, so margin was kept. Because the best setting differs between datasets, a single configuration was chosen by mean gain over Random rather than tuned per dataset.

TABLE III. Validation AULC (%) on tuning seeds 0–2 (test set unused); ✓ = chosen.

Configuration	Fashion-MNIST	SVHN	CIFAR-10	mean Δ vs Random
λ_0=0, ramp 4, margin	79.33	58.89	48.18	+2.13
λ_0=0, ramp 8, margin	79.20	58.85	47.40	+1.82
λ_0=0.5, ramp 4, margin ✓	79.75	57.53	49.19	+2.16
λ_0=0, ramp 4, entropy	79.03	58.63	47.53	+1.73
no prior (λ=1), margin	79.83	52.61	47.29	-0.09
λ_0=0, ramp 2, margin	79.28	59.04	48.07	+2.13
DualNet-AL v1 meta (reference)	79.27	54.23	47.98	+0.49
Random (reference)	78.46	53.48	48.05	–

Cost and calibration

Selection adds about one second per round on the GPU (Table IV), small next to retraining two networks. GPU k-means reduced v2's selection from 48 s to 0.03 s per round in our implementation. A weakness is calibration: DualNet-AL v2 has a higher ECE than Random, consistent with deliberately skewed training sets; temperature scaling [20] is a direct remedy we have not yet applied.

TABLE IV. Mean selection time per round (s) and mean final ECE.

Method	Fashion-MNIST (s)	SVHN (s)	CIFAR-10 (s)	mean ECE
Random	2.40	1.37	0.66	0.144
BALD [5,6]	3.00	2.14	0.82	0.187
Core-Set [7]	2.57	1.57	0.79	0.172
DualNet-AL v1 (meta)	2.85	2.22	1.08	0.227
DualNet-AL v2 (ours)	0.98	1.11	1.12	0.217

VI. Discussion and Limitations

The results support the central hypothesis that no single criterion is right for every dataset and budget, and that scheduling from typicality to learned informativeness, with cluster-level diversity, is more robust than any fixed rule. Several limitations remain. Budgets are small (≤900 labels) and pools are 10 000 images, so absolute accuracies are below full-data numbers and gaps are a few points. Three held-out seeds give a consistent paired comparison but not a formal significance test. On SVHN, v2's final accuracy (69.70%) is slightly below Random's (70.20%) even though it is ahead for almost the whole curve. The meta-learner's reward uses a 500-image validation set that the baselines do not use. Finally, the method has not yet been compared with BADGE [8] or learning-loss [9], nor evaluated on CIFAR-100 or long-tailed data.

VII. Conclusion

DualNet-AL combines two complementary networks, five acquisition signals with meta-learned weights, a budget-aware typicality prior and cluster-based batches. Under a leak-free protocol it beats random sampling on all three datasets and all held-out seeds, and is the best of nine methods on two of them. Future work covers larger budgets, more seeds with significance testing, a label-free meta reward, calibration, stronger backbones and the BADGE and learning-loss baselines.

Reproducibility. Code, per-run logs, the chosen configuration and a one-command demo are available at https://github.com/iamthehimansh/DualNet-AL_Motion.

References

[1] B. Settles, “Active learning literature survey,” Computer Sciences Technical Report 1648, University of Wisconsin–Madison, 2009.

[2] D. D. Lewis and W. A. Gale, “A sequential algorithm for training text classifiers,” in Proc. ACM SIGIR, 1994, pp. 3–12.

[3] T. Scheffer, C. Decomain, and S. Wrobel, “Active hidden Markov models for information extraction,” in Proc. Int. Symp. Intelligent Data Analysis (IDA), 2001, pp. 309–318.

[4] H. S. Seung, M. Opper, and H. Sompolinsky, “Query by committee,” in Proc. 5th Annual Workshop on Computational Learning Theory (COLT), 1992, pp. 287–294.

[5] N. Houlsby, F. Huszár, Z. Ghahramani, and M. Lengyel, “Bayesian active learning for classification and preference learning,” arXiv:1112.5745, 2011.

[6] Y. Gal, R. Islam, and Z. Ghahramani, “Deep Bayesian active learning with image data,” in Proc. ICML, 2017.

[7] O. Sener and S. Savarese, “Active learning for convolutional neural networks: A core-set approach,” in Proc. ICLR, 2018.

[8] J. T. Ash, C. Zhang, A. Krishnamurthy, J. Langford, and A. Agarwal, “Deep batch active learning by diverse, uncertain gradient lower bounds,” in Proc. ICLR, 2020.

[9] D. Yoo and I. S. Kweon, “Learning loss for active learning,” in Proc. IEEE/CVF CVPR, 2019.

[10] G. Hacohen, A. Dekel, and D. Weinshall, “Active learning on a budget: Opposite strategies suit high and low budgets,” in Proc. ICML, 2022.

[11] K. Konyushkova, R. Sznitman, and P. Fua, “Learning active learning from data,” in Proc. NeurIPS, 2017.

[12] W.-N. Hsu and H.-T. Lin, “Active learning by learning,” in Proc. AAAI, 2015.

[13] H. T. Nguyen and A. Smeulders, “Active learning using pre-clustering,” in Proc. ICML, 2004.

[14] J. Lin, “Divergence measures based on the Shannon entropy,” IEEE Trans. Information Theory, vol. 37, no. 1, pp. 145–151, 1991.

[15] R. J. Williams, “Simple statistical gradient-following algorithms for connectionist reinforcement learning,” Machine Learning, vol. 8, pp. 229–256, 1992.

[16] D. Arthur and S. Vassilvitskii, “k-means++: The advantages of careful seeding,” in Proc. ACM-SIAM SODA, 2007.

[17] H. Xiao, K. Rasul, and R. Vollgraf, “Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms,” arXiv:1708.07747, 2017.

[18] A. Krizhevsky, “Learning multiple layers of features from tiny images,” Technical Report, University of Toronto, 2009.

[19] Y. Netzer, T. Wang, A. Coates, A. Bissacco, B. Wu, and A. Y. Ng, “Reading digits in natural images with unsupervised feature learning,” in NIPS Workshop on Deep Learning and Unsupervised Feature Learning, 2011.

[20] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in Proc. ICML, 2017.

[21] A. Paszke et al., “PyTorch: An imperative style, high-performance deep learning library,” in Proc. NeurIPS, 2019.

[22] A. Blum and T. Mitchell, “Combining labeled and unlabeled data with co-training,” in Proc. COLT, 1998.